

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
12.	(a)			award (1) for each two correct A copper(II) hydroxide / $\text{Cu}(\text{OH})_2$ B ammonia / NH_3 (allow ammonium hydroxide) C $[\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}$ D copper(I) iodide / CuI E iodine / I_2 F $[\text{CuCl}_4]^{2-}$	3			3		3
	(b)			successive ionisation energies increase gradually / the energies of the <i>d</i> -orbitals are similar	1			1		
	(c)			Br_2 (1) MUST GIVE REASON TO GAIN THIS MARK bromate is reduced to bromide / bromate is a strong enough oxidising agent to oxidise VO^{2+} as it has a more positive E^0 value OR $\text{EMF} = +0.44 \text{ V}$ and this is positive so it is feasible (1) bromate is a strong enough oxidising agent to oxidise the bromide formed in the first step to Br_2 as it has a more positive E^0 value OR $\text{EMF} = +0.35 \text{ V}$ and this is positive so it is feasible (1)			1 1 1	3		
	(d)			reaction is first order (1) MUST SHOW WORKING TO GAIN THIS MARK convert any two or more pH values to $[\text{H}^+]$ (0.1000, 0.0398, 0.0158, 0.0063 mol dm^{-3} respectively) (1) show that the ratio of concentrations equals ratio of rates OR calculate $\text{rate} \div [\text{H}^+]$ for both concentrations and show it is the same (1)		1	1 1	3	1 1	

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	(e)			award (1) for appropriate unit conversion (kJ mol^{-1} to J mol^{-1} or reverse for R) rearrangement of equation gives $A = k \div e^{-(79333/RT)}$ (1) $A = 1.864 \times 10^{13}$ (1) $k = 0.167$ (1)	1		1 1 1	4	4	
	(f)			any three for (1) each - indicator colour may be difficult to see due to coloured compounds / the vanadium compounds change colour during titration - colour change of transition metals can be used in place of an indicator - for colorimetry find a wavelength OR frequency of light absorbed <u>only by the VO^{2+} / Ce^{4+}</u> - end point is when there is no absorbance at VO^{2+} frequency / <u>Ce^{4+}</u> remains in solution		2	1	3		3
				Question 12 total	5	3	9	17	6	6